

Rohan Jain

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SUMMARY

MS Data Science graduate from CU Boulder (GPA 3.9/4.0, May 2026) with production experience designing evaluation harnesses and reliability systems for an on-prem LLM agent as an AI Engineering Intern at Bright Machines, plus end-to-end ML pipeline and deep learning work at Nexus Weather and Climate. Delivered measurable results: tool-selection recall raised from 0.47 to 0.97 via a controlled study, 47% MAE reduction via Random Forest bias correction, and 90% drop in failed pipeline runs. Seeking ML Engineer roles focused on model development, evaluation rigor, and production reliability.

SKILLS

ML Frameworks & Modeling: scikit-learn, XGBoost, LightGBM, Random Forest, GBRT, SVR, MLP; regression, classification, time-series forecasting, feature engineering, hyperparameter tuning (RandomizedSearchCV), cross-validation

Deep Learning & LLM Systems: TensorFlow, Keras, PyTorch; CNN-LSTM, custom loss functions (CRPS, hybrid CRPS+MSE), probabilistic modeling, uncertainty quantification; LangGraph, LiteLLM, Ollama, RAG, embeddings (bge-m3), BM25, Model Context Protocol (MCP), LLM evaluation harnesses, Langfuse

MLOps & Pipelines: MLflow experiment tracking, reproducible training pipelines, model versioning, artifact bundles, inference quality gates, schema/unit/range validation, controlled ablation studies, mutation testing, Databricks (Unity Catalog, Model Serving, MLflow)

Languages & Cloud: Python, R, SQL (PostgreSQL, MySQL), pandas, NumPy, Apache Spark; Docker, Kubernetes, GitLab CI/CD, GitHub Actions; AWS (Lambda, S3, EC2), Azure (AKS)

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 – Present | San Francisco, CA

- Owned tool selection for an on-prem LLM agent (**LangGraph** ReAct over **LiteLLM**): 132 **MCP** tool schemas exceeded the 32K-token window; built a 115-question harness that raised recall 0.47 to 0.97.
- Killed my own recommended tool-gate design after the same harness measured it at 0.03 infra recall; shipped the conditional gate instead and stamped the correction on the original document.
- Cut ingestion cost from an estimated \$5+ per page (agent-fetched, ~280K context tokens per run) to a measured **\$0.89** by moving fetching into code, showed recurring spend tracks change rate (~12 pages a month) and rejected a re-architecture whose target spend was \$5 per year.
- Benchmarked BM25 against dense embeddings and their fusion on a real technical corpus (78% vs 83% vs 82%) and documented why fusion fails: the two retrievers err in opposite directions.
- Raised automated test coverage of a production AI orchestrator from 0 to **135 tests** and wired the suite into GitLab CI, gating every merge request for the first time.

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 – May 2026 | Boulder, CO

- Replaced legacy SVM bias-correction with a tuned Random Forest for NWP post-processing; applied feature engineering and cross-validation, reducing temperature MAE 47% and humidity MAE 54%.
- Designed a probabilistic CNN-LSTM with a custom CRPS loss for calibrated ensemble uncertainty quantification, packaging reproducible artifact bundles and achieving a best-reported CRPS of 0.1787.
- Implemented an automated inference quality gate rejecting models whose post-ML MAE fails to beat the pre-ML baseline; reduced failed production runs 90% and saved 20 engineer-hours monthly.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 – May 2026 | Boulder, CO

- Reverse-engineered proprietary Thermo Isodat binary formats via DLL decompilation, building deterministic R readers; migrated lab infrastructure to AWS EC2 with Keycloak OIDC and Grafana/Loki/Prometheus observability.

EDUCATION

University of Colorado, Boulder

Aug 2024 – May 2026 | Boulder, CO

Master's in Data Science (GPA: 3.9/4.0)

Mumbai University, Rajiv Gandhi Institute of Technology

Jul 2020 – May 2024 | Mumbai, India

B.E. in Computer Engineering (GPA: 3.7/4.0)

PROJECTS

MLflow Weather Benchmark - Experiment Tracking Pipeline | Python, scikit-learn, XGBoost, MLflow, GitHub Actions

Jun 2026

- Reproducible benchmarking of 5 regression models on live Open-Meteo data with every run logged to MLflow; an automated quality gate rejects models that fail to beat a baseline (best: 84.8% MAE improvement), CI smoke test on every push.

Yoga Posture Detection & Correction | Python, TensorFlow, Keras, MediaPipe

Jan - May 2024

- Built a two-stage CV pipeline: MediaPipe pose estimation extracts keypoints; a TensorFlow/Keras network classifies postures with 96.5% accuracy - published as a peer-reviewed paper.